Predictive Modeling

Regression Modeling of Citigroup Daily Stocks Using S&P 500 Financial Sector Components

Abstract

The purpose of this project was to analyze daily stock prices for CitiGroup and other Financial sector companies in the S&P 500 to determine which stocks best predict CitiGroup's daily closing price. Using multiple linear regression, this analysis examines how the prices of firms such as Bank of America, Capital One, Black Rock and Bank of New York Mellon explain variation in CitiGroup's stock performance. The goal was to identify the most statistically effective model and to test whether adding interaction or quadratic terms could improve predictive accuracy.

This analysis followed a structured, multi-stage approach. Data for all financial sector companies in the S&P 500 was imported into SAS for statistical evaluation. An initial Pearson correlation analysis was used to identify the stocks most strongly associated with CitiGroup and was noted to carry forward for further analysis, followed by selecting 19 variables through stepwise regression. Exploratory Data Analysis (EDA) was then performed in Python to examine trends, detect outliers, visualize sector patterns, and understand how the selected variables behaved over time. When visualizing all 19 stock trends together, it became difficult to clearly identify patterns as they had no order to them, so the variables were grouped together according to their sectors, which made the similarities and differences in movement much easier to interpret. This process also revealed that the insurance sector stocks did not follow CitiGroup's trend as closely as the other sectors, and further research showed that insurance companies behave differently due to their unique business models, which means they are not the best predictors for CitiGroup in this context, despite their good correlation with CitiGroup. Going with 12 variables, multiple regression models were developed and refined in SAS, with variable selection based on stepwise

regression, multicollinearity diagnostics (VIF), and statistical significance tests, after which I ended up with 8 significant variables. Interaction and quadratic terms were then tested using PROC GLM and PROC PLM.

The analysis concluded that the best overall model included Bank of America, Capital One, BlackRock, and Bank of New York Mellon, along with several significant interaction terms (including BAC×COF, BAC×BK, BAC×BLK, COF×BK, and BK×BLK) and the quadratic term for Bank of New York Mellon. This complete model provided the most accurate prediction of CitiGroup's stock price, explained approximately 98.3% of its variation, and showed low prediction error. Overall, the results show the connection among major financial sector stocks and how a carefully selected set of companies can reliably predict CitiGroup's price during the 2024-2025 period.

Introduction

Stock prices play a major role in determining the financial position and market perception of companies. Understanding the factors that influence these prices is important for analysts, investors, and anyone interested in how different stocks behave. CitiGroup, as one of the largest established banking institutions in the United States, serves as an ideal case study for examining how companies within the Financial sector interact in the market. This analysis uses daily stock price data for CitiGroup and other financial sector companies listed in the S&P 500.

The dataset initially contained 250 observations representing each trading day from November 2024 to October 2025. Along with CitiGroup's price, the dataset included 75 variables with many major company stocks. From this extensive dataset, a more focused collection of predictors was developed through a series of filtering and variable selection techniques. This process allowed the model to focus on stocks that could predict CitiGroup's daily price movements.

The primary goal of this analysis is to determine the best multiple regression model for predicting CitiGroup's daily closing price using other Financial sector stocks as independent variables. Citigroup's price serves as the dependent variable, while the independent variables consist of daily closing prices for companies across several financial industry groups, including investment banking, diversified banks, consumer finance, asset management, financial exchanges, and insurance. This report will detail the process of variable selection through correlation analysis, Exploratory Data Analysis (EDA), and stepwise regression, followed by building and evaluating multiple regression models to determine which financial companies are strong predictors of CitiGroup's prices.

Methodology

The analysis began by importing the dataset of daily closing prices of CitiGroup and all Financial sector components of the S&P 500 into SAS. The original dataset contained 250 daily observations and 75 variables, covering the period from November 2024 to October 2025. Once the general structure of the data was understood, a preliminary correlation analysis, using PROC CORR was conducted to determine which financial stocks had the strongest linear relationships with CitiGroup's price. A simple Pearson correlation was run between Citigroup and all numeric variables, after which I noted down every stock that had a correlation of 0.60 and higher with CitiGroup, resulting in a list of 40 stocks. These variables were selected for the next stage of analysis.

Since 40 predictors were still too many to include in a regression model, PROC REG with selection as stepwise regression was used as a variable selection method. Stepwise testing was performed in SAS, which reduced the predictor set to 19 variables. A second correlation check and scatterplot review were completed on these 19 variables using PROC CORR with scatterplot options, to confirm their relationships with CitiGroup and to visually assess general patterns and linearity.

At this point, Exploratory Data Analysis (EDA) was performed in Python to better understand the behavior of these 19 variables. I especially wanted to inspect two things through EDA: signs of multicollinearity and whether Insurance stocks followed similar trends to CitiGroup. Using Python libraries such as Pandas, Matplotlib, and Seaborn, several visualizations were created, including time series trend plots, histograms, boxplots, and a correlation heatmap. The grouping of variables by industry was also performed in Python to identify sector-specific patterns.

Descriptive statistics, histograms and boxplots were used to check for skewness and potential outliers, while the heatmap was used to show correlations among variables. This EDA step helped confirm which variables behaved similarly, which sectors moved together, and how strongly each variable was associated with CitiGroup before modeling. Based on these trends, the Insurance sector and one stock from the Financial Exchange and Data group were removed, leaving 11 variables to carry forward.

A final stepwise regression on these 12 predictors was then run in SAS using PROC REG with stepwise regression selection, which identified 8 significant variables. At this point, multicollinearity was tested using PROC REG with the VIF to check: Variance Inflation Factors (VIF), t-tests, coefficient signs, and correlations among predictors. This step allowed me to reduce the model to the final first-order model consisting of three companies.

After finalizing the first order model, possible interaction effects were tested using PROC GLM and PROC PLM in SAS. Interaction plots were created in SAS using effectplot slicefit to check whether the relationship between CitiGroup's price and one predictor changed at different levels of another predictor. Next quadratic terms were tested for each predictor in PROC REG to assess potential non linear relationships.

Finally, residual diagnostics including residual versus predicted plots, Q-Q plots, residual distributions and Cook's D, generated from PROC REG (with all diagnostics unpacked) were reviewed to verify that the regression assumptions were satisfied.

Correlation Analysis

The first step in the model building process was to identify which financial sector stocks had a strong linear relationship with the dependent variable, CitiGroup's daily closing price (C) . A simple Pearson correlation was computed between C and all numeric variables in the dataset, which consisted of price series for 74 Financials industry components and CitiGroup. The goal of this step was to narrow the initial list of predictors to only those variables with good explanatory potential.

The initial correlation screening revealed that 40 variables exhibited strong linear relationships with CitiGroup, specifically selecting the ones with correlation coefficient of $|r| >= 0.60$. Setting this as the threshold allowed me to only look at stocks whose movements align closely with Citi's stock. The large number of strongly correlated variables (40 variables) from initial screening were expected, as financial stocks often move together due to market-wide economic forces.

To further reduce the dimensionality of the model before proceeding to full exploratory data analysis, a stepwise regression was performed. The stepwise selection procedure reduced the predictor set from 40 variables down to 19 variables.

Once these 19 variables were identified, a second correlation analysis was conducted, this time focusing exclusively on the reduced set of predictors. Pearson correlation coefficients and p-values were examined, and this time scatterplots of Citi's price versus each of the 18 predictors were generated to visually assess the linearity of the relationships. This step allowed me to get a clearer picture, limiting the analysis to 18 strongly associated variables for more interpretable plots and reduced visual clutter.

The results of the correlation analysis of these 19 variables, performed using PROC CORR in SAS, are summarized in **Table 1**.

Independent Variable	**Correlation Coefficient, r**	**P-value**	**Reject H0: beta1 = 0**
GS	0.984	<0.0001	Yes
JPM	0.977	<0.0001	Yes
BK	0.967	<0.0001	Yes
IBKR	0.964	<0.0001	Yes
MS	0.947	<0.0001	Yes
HOOD	0.945	<0.0001	Yes
COF	0.9403	<0.0001	Yes

NDAQ	0.886	<0.0001	Yes
BAC	0.882	<0.0001	Yes
BLK	0.8785	<0.0001	Yes
BEN	0.846	<0.0001	Yes
AXP	0.843	<0.0001	Yes
SYF	0.832	<0.0001	Yes
GL	0.808	<0.0001	Yes
ERIE	-0.796	<0.0001	Yes
MCO	0.6565	<0.0001	Yes
HIG	0.655	<0.0001	Yes
TRV	0.605	<0.0001	Yes
WRB	0.603	<0.0001	Yes

Table 1: Correlation Analysis of Predictor Variables with CitiGroup (C)

As shown in the table, the correlation revealed extremely strong relationships between CitiGroup and several major Financial stocks, including GS, JPM, MS, IBKR, BK, COF, BAC, BLK, BEN, AXP, SYF, GL, and ERIE; with few of these stocks above 0.95.

The scatterplots in **Figure 1** visually confirm the nature and strength of the linear relationships for the first 10 scatter plots. Each of the highly correlated predictors showed a tight, upward sloping linear pattern with minimal dispersion around the trend. The scatter plots for GS, JPM, BK, IBKR, MS, HOOD, COF, NDAQ, BAC, and BEN look very similar to each other, each showing Citi's price increasing almost proportionally as the predictor variable increased.

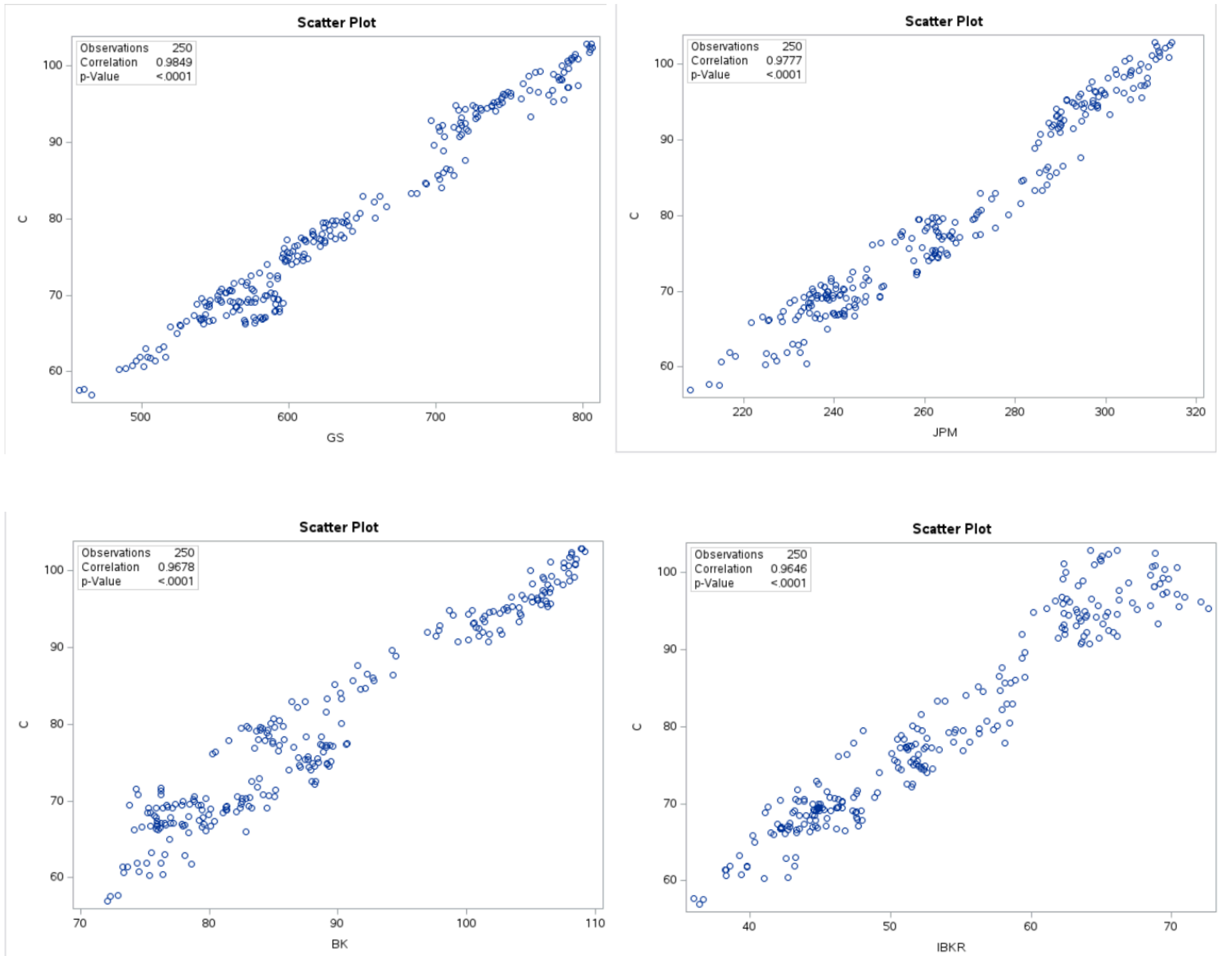


Figure 1: Correlation Plots of GS, JPM, BK, AND IBKR with CitiGroup (C)

The four plots above show the tightest and most well-defined linear pattern, clearly illustrating how CitiGroup's stock price increases almost proportionally as GS, JPM, BK, and IBKR increase. Each predictor exhibits a strong positive linear association with C, reflected by their high correlation coefficients shown in **Figure 1.** These relationships indicate that these stocks tend to move closely together within the Financials sector, responding similarly to broad market conditions and systemic economic forces.

The scatterplots for the remaining predictors, HOOD, COF, NDAQ, BAC, BLK, BEN, AXP, and SYF, display the same overall structure and visual characteristics. Each of these variables forms a similarly tight, upward-sloping linear pattern with minimal variance around the fitted line, confirming the strength and consistency of their relationships with CitiGroup. Because these additional scatterplots do not provide new or distinct visual insights, they have been included in the **Appendix**.

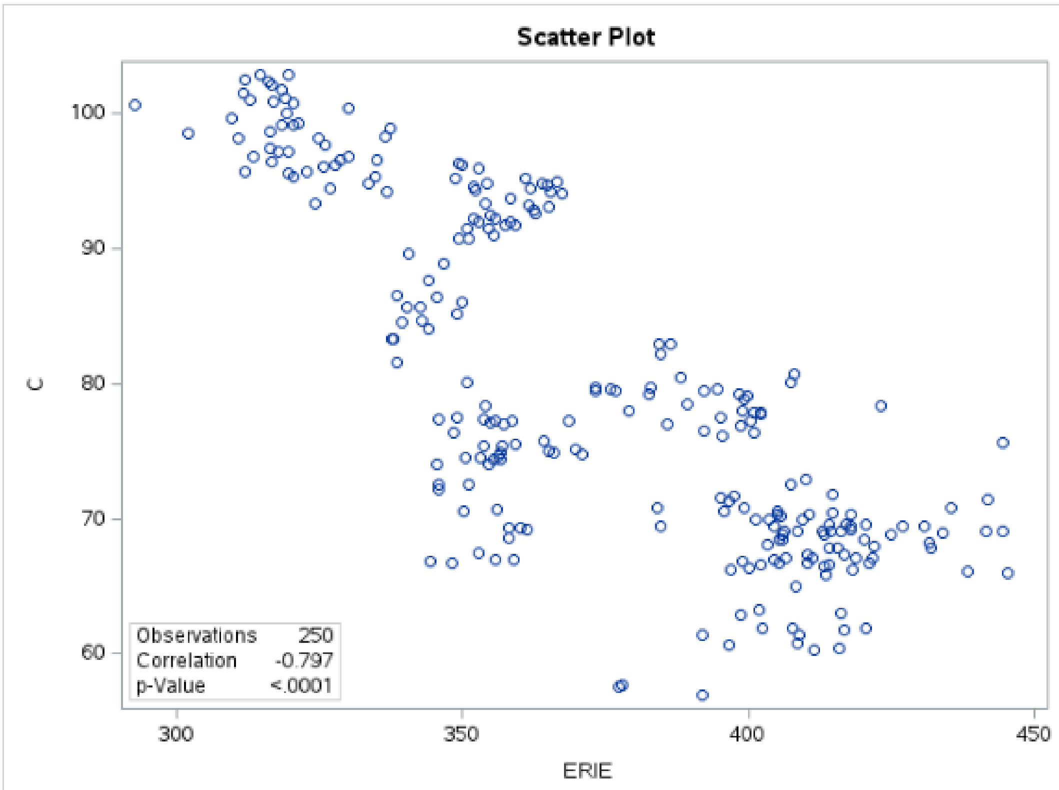


Figure 2: Correlation Plot of ERIE with CitiGroup (C)

In addition to the strongly linear relationships shown in **Figure 1,** the scatterplot for ERIE (**Figure 2)** shows a strong negative correlation. ERIE shows a statistically significant linear association with CitiGroup (r = -0.797), the direction of the trend slopes downward, indicating that ERIE's stock price tends to move inversely relative to Citi's.

The scatterplots for the remaining variables, GL, MCO, HIG, TRIV, and WRB, were also examined as part of the exploratory analysis. Although scatterplots of each of these predictors displayed similar linear patterns as **Figure 1,** they had more dispersion, particularly the WRB plot (**Appendix).**

The correlation and scatterplot analysis provided a clear understanding of the strength and direction of the linear relationships between CitiGroup and the selected financial-sector predictors. The next step is to closely examine the underlying characteristics of the variables through Exploratory Data Analysis.

Exploratory Data Analysis

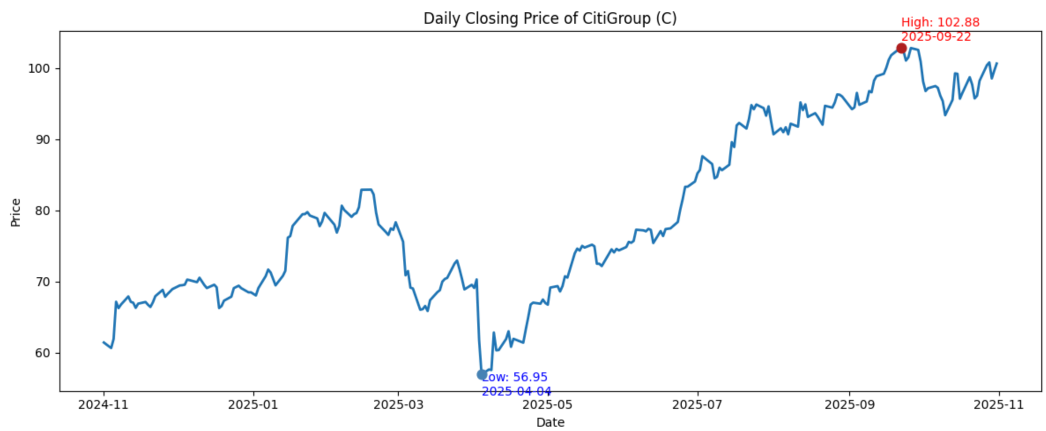
The purpose of the Exploratory Data Analysis (EDA) is to better understand the characteristics and behavior of the variables selected for this analysis before building the regression models. Since the dataset consists of daily stock prices, it is important to examine the distributions, identify any outliers, and observe general trends over time. EDA also provides insight into how the selected financial stocks behave relative to CitiGroup.

A numerical summary provides the foundation for this analysis. **Table 2** displays the key descriptive statistics for the 18 variables in the dataset. These statistics provide an initial overview of the center, spread, and general behavior of each stock's daily closing price.

	count	mean	std	min	25%	50%	75%	max	skew	kurtosis
C	250.0	79.728156	12.467603	56.952473	69.152731	77.149578	92.778685	102.876602	0.336795	-1.215507
GS	250.0	639.336983	90.099021	457.478516	571.262177	614.293213	719.187607	806.320007	0.300520	-1.058533
JPM	250.0	265.268797	27.812136	208.259308	239.868546	262.402985	290.447906	314.530823	0.135238	-1.253456
BK	250.0	89.010309	11.252742	72.116707	79.375816	86.493813	100.872496	109.211365	0.417446	-1.209764
IBKR	250.0	53.353773	9.414219	35.971451	44.863009	51.791359	62.387898	72.620003	0.254784	-1.172303
MS	250.0	132.792696	15.131604	97.756958	123.314554	130.296288	142.054771	165.015411	0.189150	-0.539993
HOOD	250.0	72.804680	35.836759	23.969999	41.944999	56.850000	104.262497	152.460007	0.626339	-0.986109
COF	250.0	196.515605	19.012918	149.683548	182.014580	196.486069	214.349995	229.740005	-0.166984	-0.916991
NDAQ	250.0	83.084694	7.218962	66.007294	77.702126	81.321182	88.812315	96.667015	0.180885	-0.894308
BAC	250.0	45.308580	3.976890	33.999416	43.357644	45.367357	47.677310	53.450001	-0.276777	0.059819
BEN	250.0	21.256167	2.355729	16.207933	19.269042	21.023619	23.516030	25.673576	0.136130	-1.161531
AXP	250.0	300.244144	25.521816	230.223938	285.509575	299.241318	316.284500	361.670013	-0.107730	0.176754
GL	250.0	123.855474	11.189018	102.627335	118.645807	121.675495	134.277496	146.266235	0.133947	-0.844136
ERIE	250.0	372.938433	36.164549	292.640015	348.309349	365.195404	405.317825	445.350189	-0.020036	-1.093636
MCO	250.0	481.221747	23.970732	395.220917	469.508339	482.245010	498.596977	524.527222	-0.845924	1.078905
HIG	250.0	120.969038	7.935768	105.004417	114.816086	121.777092	127.647552	134.699997	-0.153312	-1.136005
TRV	250.0	258.212307	12.855560	229.605042	248.013176	259.576935	268.587494	284.970001	-0.255364	-0.840072
WRB	250.0	67.095803	6.260455	56.106182	60.838592	69.086174	72.198856	78.459999	-0.228488	-1.372570
BLK	250.0	1022.037241	86.586188	807.609863	960.598587	1008.211243	1104.796509	1202.589966	0.013165	-0.803565

Table 2: Descriptive Statistics of Key Predictor Variables

The statistics show that there is considerable variation in price levels across the different financial companies. Citigroup has an average daily closing price of about 79.7 with low variability (standard deviation = 12.47). Several firms, such as GS, BLK, MCO, ERIE, and AXP have much higher price levels and wider spreads, while BK, BAC, and BEN show lower average prices. Most variables exhibit mild skewness and slightly negative kurtosis, indicating fairly symmetric distributions without extreme outliers.



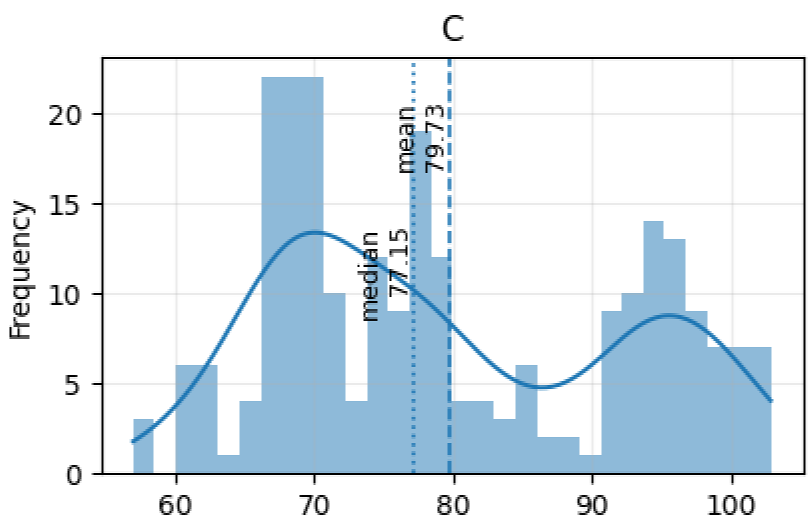
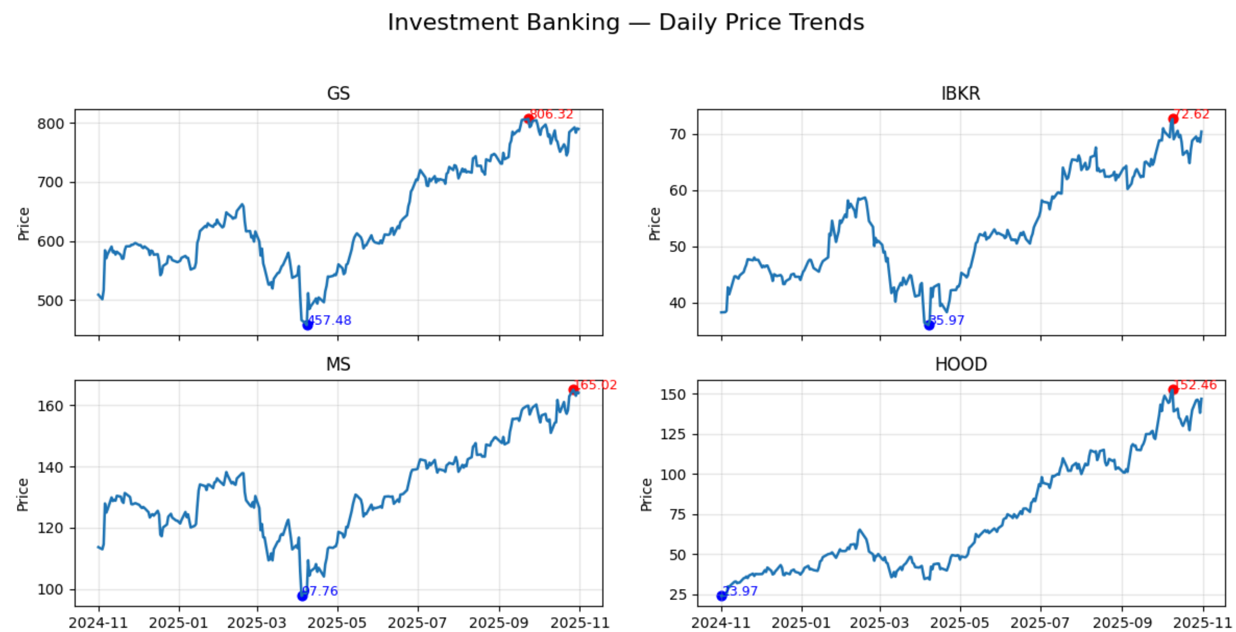


Figure 2: Daily Closing Price of CitiGroup (C)

Figure 2 shows the daily closing price of Citigroup over the period from November 2024 to October 2025. The stock exhibits short-term fluctuations but a clear upward trend across the year. Citi reached its lowest price of $56.95 on April 4, 2025, during a period of sustained decline in early spring. After this point, the stock recovered steadily and continued rising until October, reaching its highest closing price of $102.88 on September 22.

To understand the behavior of the 18 selected financial stocks, they were grouped according to their industry classification within the financial sector. This grouping allowed me to compare them easily with similar business models and market drivers. The variables were grouped into these 6 categories: Investment Banking (GS, IBKR, MS, HOOD), Diversified Banks (JPM, BAC), Consumer Finance (COF, AXP), Asset Management (BEN, BK), Financial Exchange and Data (NDAQ, MCO), and Insurance (HIG, TRV, WRB, ERIE). Organizing the stocks this way made it easier to look at sector specific trends, and identify common price movement patterns. The following figures show the daily closing prices for each sector grouping over the one year period.



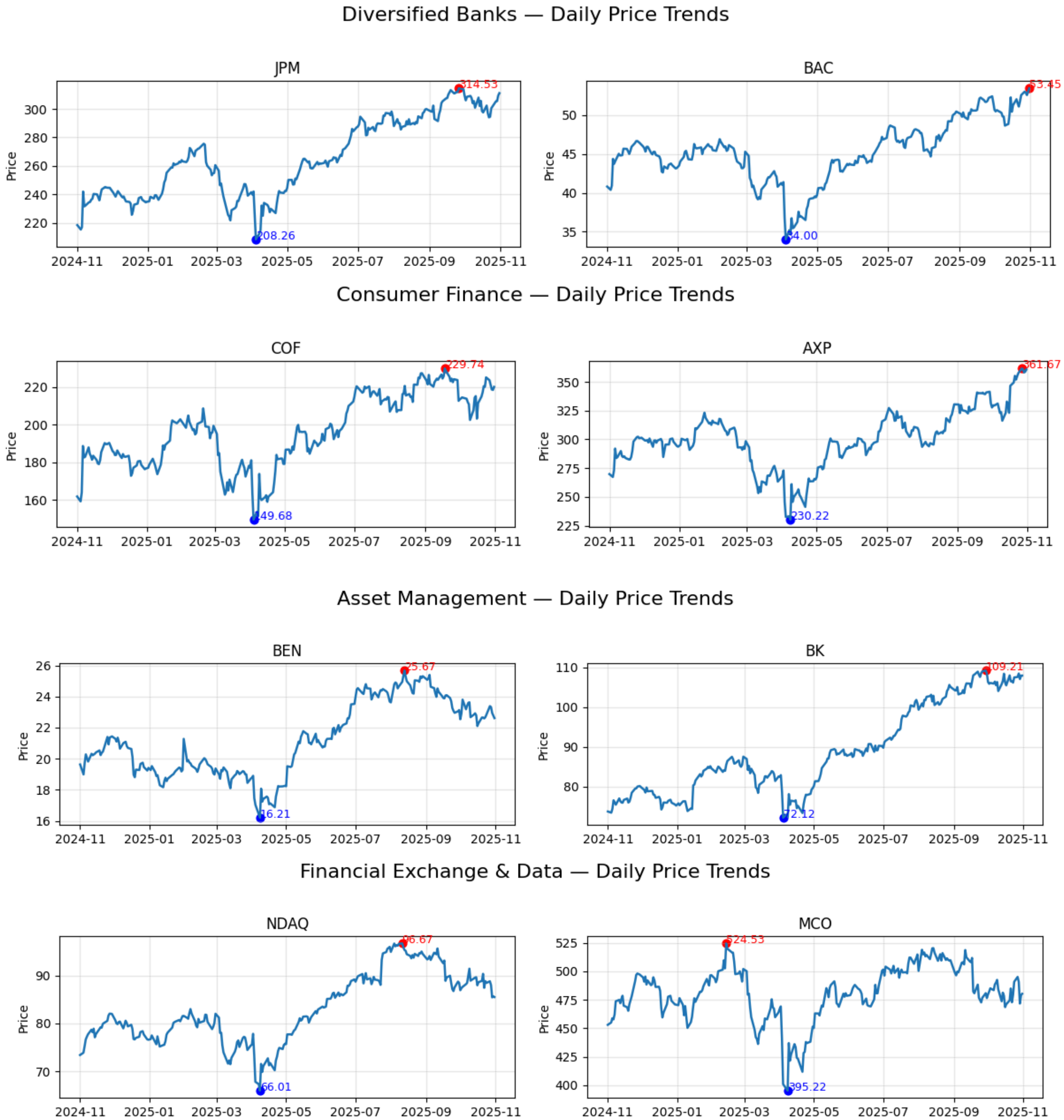


Figure 3: Daily Closing Prices of Stocks

When looking at the plots in **Figure 3,** one pattern stood out right away: most of these stocks move in almost the same direction at the same time. In particular, stocks such as GS, IBKR, MS, JPM, BAC, COF, AXP, BEN, BLK, BK, and NDAQ all follow very similar trend paths. They rise and fall at nearly the same points in time, showing their lowest prices in April 2025 and reaching their highest levels around October 2025. This also aligns with the trend line of CitiGroup.

This similarity indicates that any impact to the financial sector affects most companies in that sector at the same time. The fact that so many of these stocks share almost identical turning points in the year suggests general market forces. This could also explain why the correlation analysis showed such high correlations across these variables.

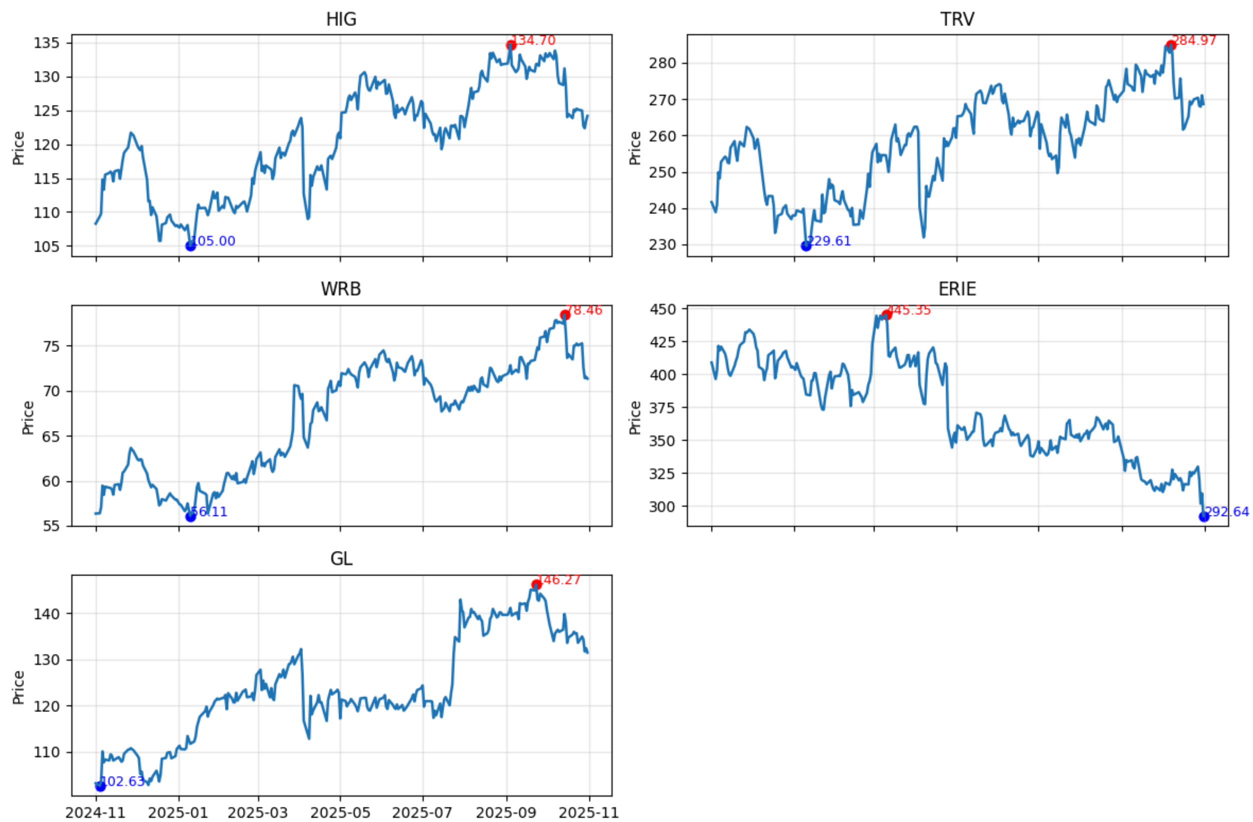
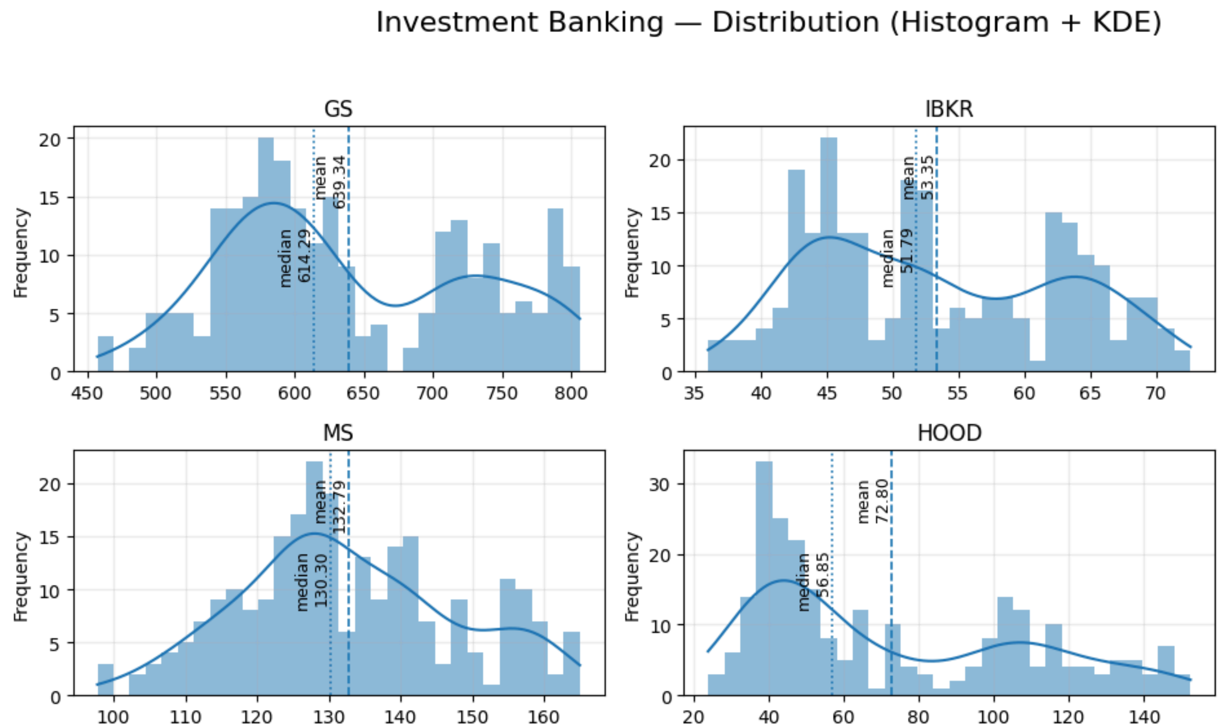
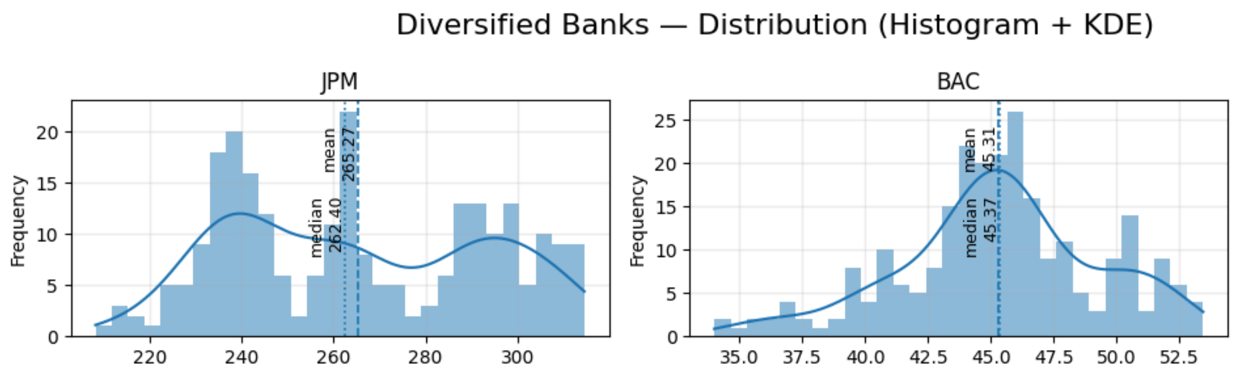
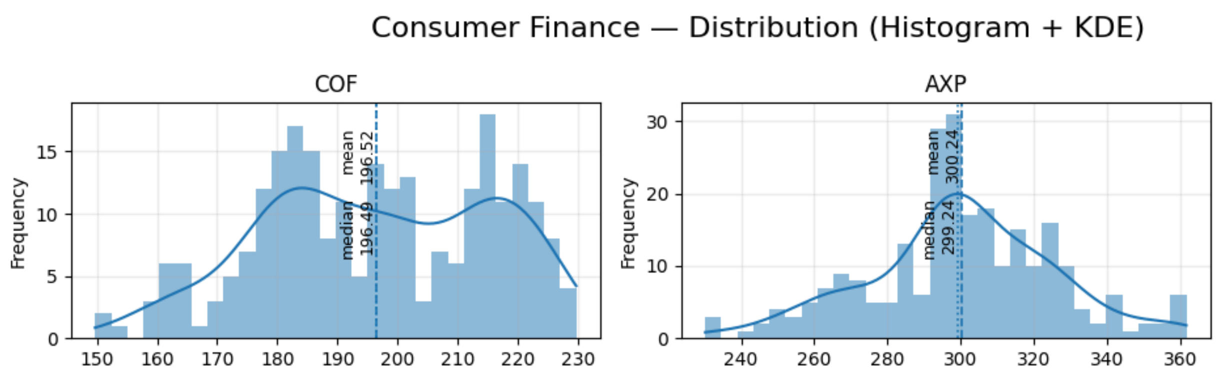


Figure 4: Daily Closing Prices of Insurance Stocks

The insurance stocks show more variation in trend behavior compared to the other financial groups. HIG and TRV seem to have an upward trend with some dips in the middle and WRB also shows trends upwards overall, with some short term fluctuations. In contrast, ERIE moves in the opposite direction, declining consistently over the year and ending near its lowest point in late October. GL shows moderate growth with a noticeable rise early in the year and at the end of the year.


Investment Banking — Distribution (Histogram + KDE)
GS
mean
639.34
median
614.29
IBKR
mean
53.35
median
51.79
MS
mean
132.79
median
130.30
HOOD
mean
72.80
median
56.85
Frequency


Diversified Banks — Distribution (Histogram + KDE)
JPM
mean
265.27
median
262.40
BAC
mean
45.31
median
45.37
Frequency


Consumer Finance — Distribution (Histogram + KDE)
COF
mean
196.52
median
196.49
AXP
mean
300.24
median
299.24
Frequency

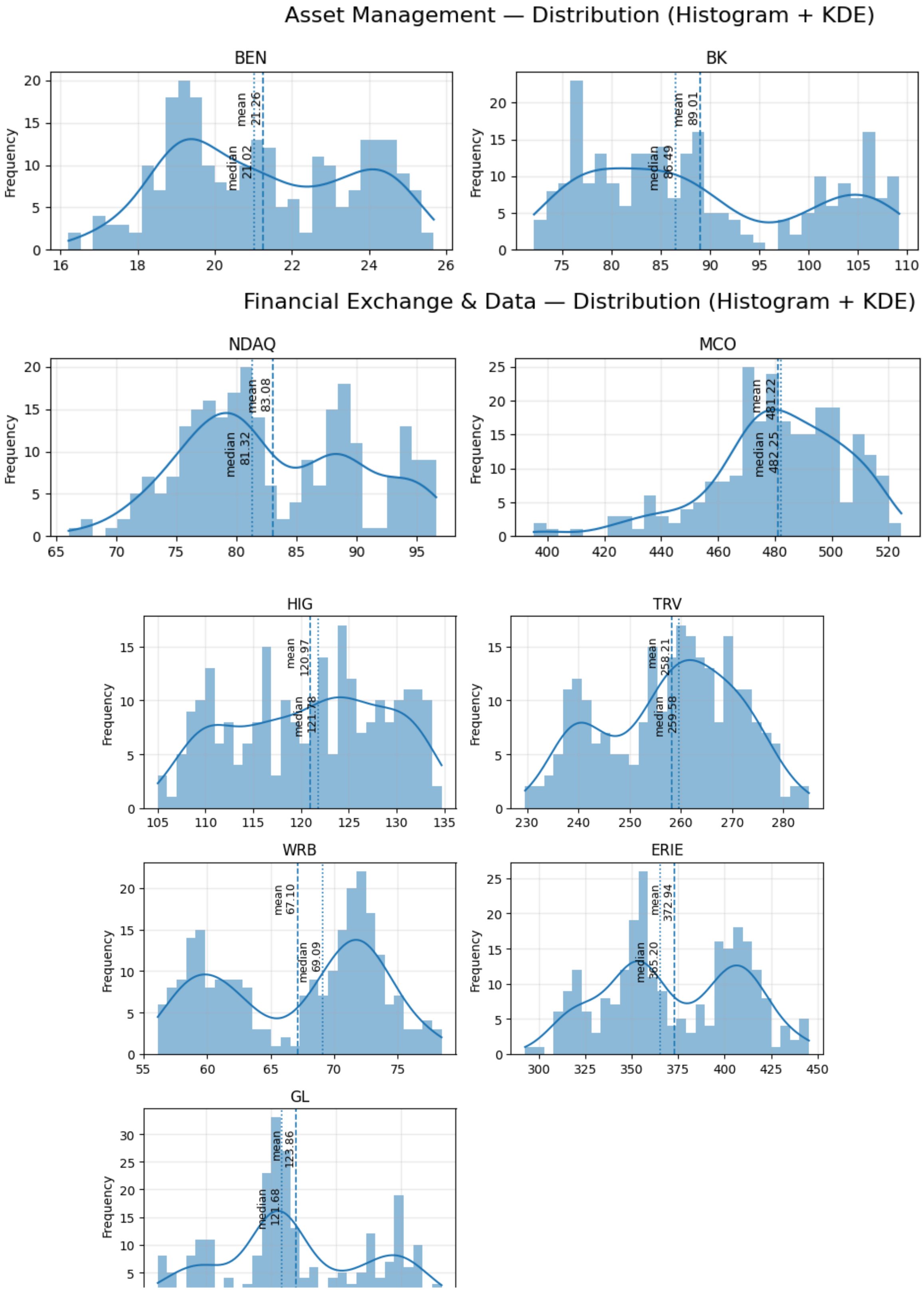


Figure 5: Distribution of Daily Closing Prices

The distribution plots show that the daily closing prices for most of the stocks are roughly bimodal, with a cluster of values that reflects each company's typical trading range over the year. No stock shows extreme outliers or unusual shapes. Overall, these distributions suggest that the daily price movements for the selected stocks are relatively stable, supporting their suitability for regression modeling

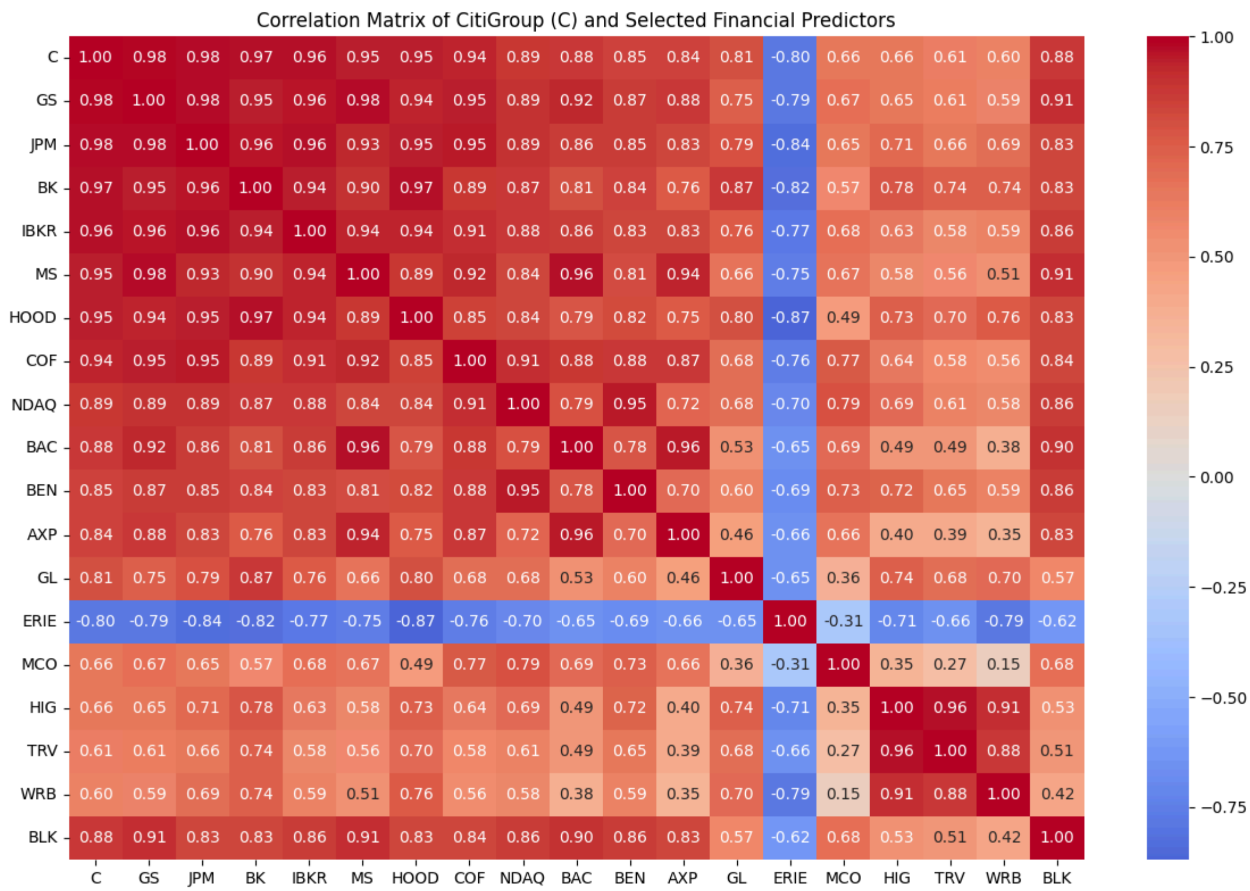


Figure 6: Correlation Heatmap of Predictor Variables

The correlation matrix shows extremely strong positive relationships among almost all of the financial sector stocks. Many of the predictors including GS, JPM, BK, IBKR, MS, COF, BAC, and NDAQ have correlations above 0.90 with CitiGroup, reflecting the fact that these companies tend to move together in response to broad market and economic conditions. Another clear pattern is the strong correlation within sectors: investment banking stocks are highly correlated with each other, diversified banks move closely together, and insurance companies form their own clusters. This shows that companies within the same sector share similar risk exposures and are influenced by the same things. ERIE behaves differently from the rest of the financial sector during this period. The heatmap confirms that multicollinearity is expected when modeling Citi's price with these predictors, since many variables may contain overlapping information. This will need to be addressed next.

Based on the Exploratory analysis, the Insurance Groups (HIG, TRV, WRB, ERIE, GL) and MCO will be removed from consideration for regression modeling. There are 2 reasons behind doing so, firstly insurance companies and financial exchanges operate under different models, and their stock prices may respond to different forces than those affecting CitiGroup. This was reflected in the trend plots: while most banking and consumer finance stocks rose and fell together throughout the year, the insurance stocks exhibited more irregular movement.

The correlations had also confirmed that the insurance stocks and MCO had lower correlations to Citi. This behavior indicates that insurance industry price movements do not align closely enough with Citi's daily price changes to serve as reliable predictors in a regression model focused on CitiGroup.

Hence, the Insurance group and MCO were removed from the analysis, leaving a final set of 12 variables to be carried forward into the regression modeling stage; these 12 variables are: GS, IBKR, MS, HOOD, JPM, BAC, COF, AXP, BEN, BK, BLK and NDAQ.

Multiple Regression Modeling

Analysis of the 1st Order Model

Following the Exploratory Data analysis, the next step was to develop a multiple regression model to predict CitiGroup's daily closing price (C) using the selected financial sector variables. Because the dataset still contains 12 variables, stepwise regression was used as the primary variable selection technique. This method evaluated predictors one at a time and added them to the model when they provided a statistically significant improvement in explaining the dependent variable.

Statistic	Value	Interpretation
R-Square	0.9887	Explains 98.8% variation
Adj R-Square	0.9881	High explanatory power
BIC	177.4827	Lowest BIC
C(p)	9.05	Low C(p), good
PRESS	480.59	Lowest PRESS
P-value	<0.05 (all)	All predictors are significant

Table 3: 1st Order Model from Stepwise Selection

The stepwise procedure identified 8 variables as statistically significant predictors for CitiGroup: GS, BK, COF, BEN, MS, BLK, BAC, and IBKR; removing AXP, HOOD, JPM, and NDAQ. GS first entered the model with an R-square of 0.97, which means it accounted for approximately 97% of the variation in C. Each additional predictor contributed a smaller but meaningful improvement to the model, and the final first-order model achieved an R-square of 0.9887 and C(p) of 9.05, indicating that the selected variables collectively explain nearly all variation in the daily closing price of CitiGroup. All variables are significant since their p-values are less than 0.05.

Even though all predictors in the stepwise model are statistically significant at the 0.05 level, stepwise regression alone does not evaluate whether the predictors are redundant or overlapping in the information they provide. The next step is to check for multicollinearity as it was suspected in the earlier examination of the variables given their extremely high correlations between variables.

Examining Multicollinearity

After obtaining the nine variable first-order model from stepwise selection, multicollinearity diagnostics were conducted to check whether any predictors were providing redundant information. Variance inflation factors (VIFs), t-tests, coefficient sign changes, and pairwise correlations were examined to identify instability within the model. Several predictors exhibited extremely high VIF values (>10), confirming the presence of multicollinearity. Following the recommended procedure, the predictor with the highest VIF was removed first, and the model was re-estimated. This process was repeated iteratively, each time eliminating the variable with high VIF. BEN was removed even though it was statistically significant and had low VIF, because it had a negative regression coefficient, which contradicted its positive correlation with CitiGroup and its upward trend in the time-series plots in the EDA. A sign reversal of this kind is an indication of multicollinearity.

The final reduced model contained three predictors: BAC, COF, BLK, and BK, all of which showed strong statistical significance and VIF values well below the threshold of 10.

Metric	**Model 1: BAC + COF + BK + BLK**
T-test (P-value)	<0.0001 <0.0001 <0.0001 0.0020
Adj R-Square	0.9746
Root MSE (s)	1.9866
Coefficient of Variation	2.49 %
F-test (P-value)	<0.0001

Table 4: 4-Variable Multiple Regression Model Statistics

The four variable first-order model is statistically strong. All four predictors are significant at the 0.05 level, the adjusted R-square shows that the model explains 97.4% of the variation in CitiGroup's daily closing price, and the Root MSE is relatively small compared to the scale of the dependent variable. In addition, the coefficient of variation is only 2.5%, indicating that the model has a low prediction error. The overall F-test is also significant.

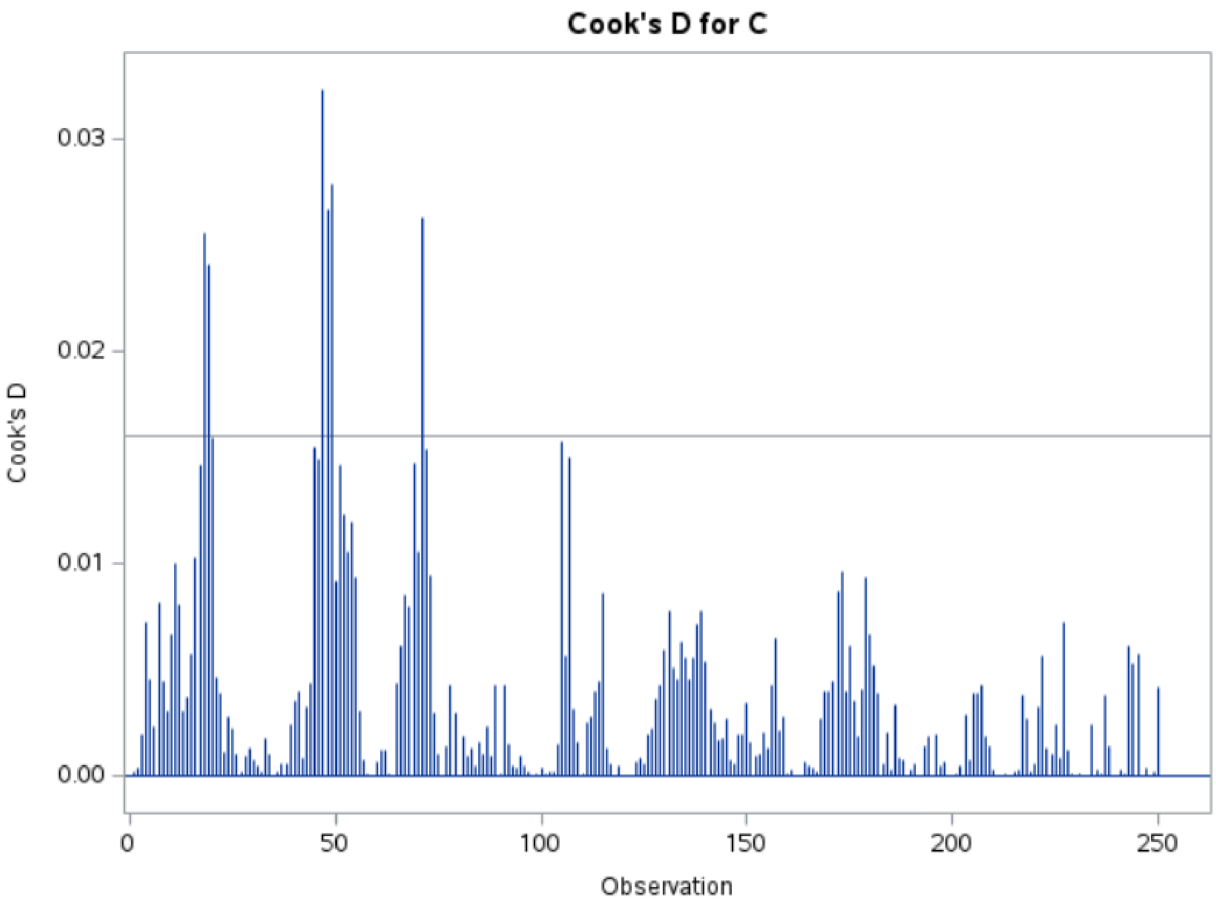


Figure 7: Cook's D Plot

The studentized residuals did not reveal any extreme outliers, and the majority of the observations fell within the expected range. A few points showed slightly larger residuals, but their Cook's D values weren't too off the threshold, indicating that no single observation had an undue influence on the fitted model.

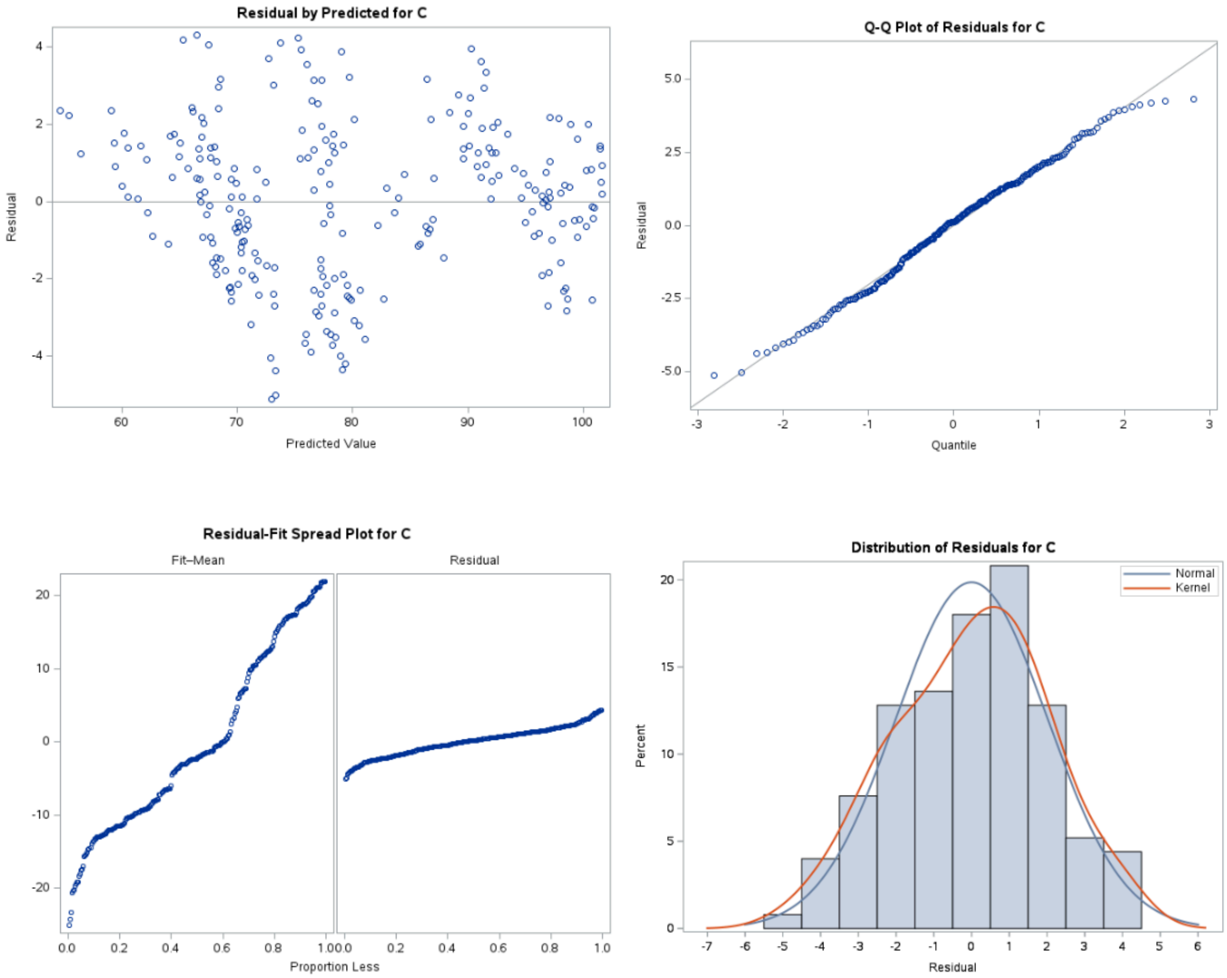


Figure 8: Diagnostic Plots

The residual versus predicted plot showed a good random scatter of plots centered around zero, with no visible curvature or pattern. The Q-Q plot shows that points closely follow the reference line, indicating that the residuals were normally distributed. The residual fit spread plots also showed a symmetric and well behaved pattern (fit mean is taller than residual). And the histogram of residuals closely resembles a normal distribution with only minor deviations in the tail.

Looking at all the diagnostics, this three variable first order model appears to be statistically significant and appropriate for predicting CitiGroup's daily closing price.

Adding Interaction and Second-Order Terms to the Best Model

After building the three variable first-order model, the next step was to check whether any interaction terms were needed. Interaction plots were made for all possible pairs of predictors to see if the relationship between one variable and CitiGroup's price changed at different levels of another variable. In every plot, the lines were not parallel and had points of intersection **(Appendix)**. This visual evidence indicated that interactions might be important.

Each interaction term was then added to the regression model. The statistical tests showed that almost all interaction terms were significant, with p-values below 0.05, meaning they contributed meaningful explanatory value to the model. The only interaction that was not strongly significant was COF_BLK, which had a higher p-value. All other interaction terms were statistically significant and improved the model's performance.

Metric	**Model 1:** **BAC + COF + BK + BLK**	**Model 2 (Interaction):** **BAC + COF + BK + BLK + BAC_COF + BAC_BK + BAC_BLK + COF_BK + BK_BLK**	**Model 2 (Complete):** **BAC + COF + BK + BLK + BAC_COF + BAC_BK + BAC_BLK + COF_BK + BK_BLK + BK_SQ**
T-test (P-value)	<0.0001 <0.0001 <0.0001 0.0020	<0.0001 0.0280 <0.0001 0.0007 <0.0001 0.0292 <0.0001 <0.0001 <0.0001	<.0001 0.0271 <.0001 0.0034 0.0002 0.0013 <.0001 0.0219 <0.0001 0.0183
Adj R-Square	0.9746	0.9831	0.9834
Root MSE	1.986	1.619	1.604
Coeff Var	2.491 %	2.031 %	2.012%
F-test	<0.0001	<0.0001	<0.0001

Table 5: Comparison between Models

After adding the significant interaction terms, the adjusted R-square increased from 0.9746 to 0.9831, and the Root MSE decreased from 1.9866 to 1.619, showing an improvement in predictive accuracy.

Next, quadratic terms for BAC, COF, BLK, and BK were tested to check for potential nonlinear relationships. Only the squared term for Bank of New York Mellon (BK_SQ) was statistically significant. Adding BK_SQ to the interaction model further improved the model fit, increasing the adjusted R-square to 0.9834 and lowering the Root MSE to 1.604.

Based on these improvements, the complete model including the significant interaction terms and the BK_SQ quadratic term was selected as the best overall model.

Analysis of the Overall Best Model:

1. Significance Tests: The ANOVA F-test ($p<.0001$) confirms the overall model is statistically significant, which includes the first order terms, interaction terms, and one quadratic term. All predictors in the final model including Bank of America (BAC), Capital One (COF), Bank of New York Mellon (BK), BlackRock (BLK), their significant interaction terms, and the squared term for BK have p-values below 0.05, meaning they all make meaningful contributions to predicting CitiGroup's stock price.
2. Model Fit (R-Square): The final model has an Adjusted R-Square of 0.9834, meaning the predictors explain about 98.3% of the CitiGroup's daily closing price. This is an extremely strong fit for financial data.
3. Prediction Error (Root MSE and Coefficient of Variance): The standard deviation of the error, or Root MSE, is 1.604, which means we expect 95% of the predicted CitiGroup prices to fall within roughly ±3.21 (2 × Root MSE) of the regression line. The Coefficient of Variation is 2.012%, which is well below the commonly used 10% benchmark, indicating that the model has very low relative prediction error and provides highly accurate predictions.

Diagnostics Plot

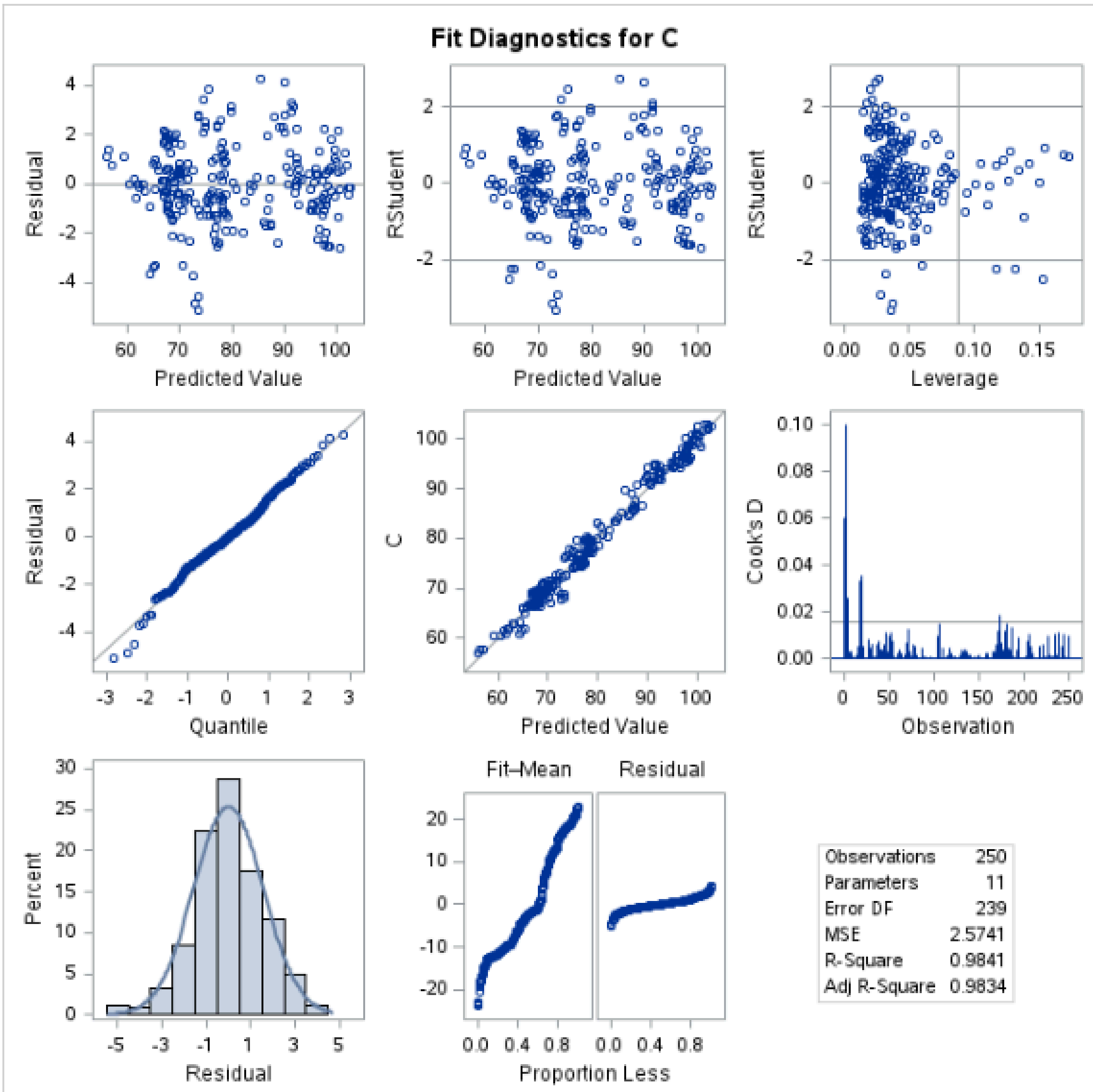


Figure 9: Diagnostic Plots

Several diagnostic plots were used to check whether the assumptions of the regression model were satisfied. The residual versus predicted plot showed a random scatter of points centered around zero, with no clear pattern or curved shape. This suggests that the model's assumptions of linearity and constant variance are reasonably met.

The Q-Q plot displayed points that closely followed the reference line, indicating that the residuals are approximately normally distributed. The Fit-Mean Residual plot also showed a symmetric pattern without major deviations, supporting the idea that the model fits the data well across the full range of predicted values.The residual distribution plot aligns closely with the normal curve, confirming that the residuals follow a roughly normal distribution with no extreme outliers.

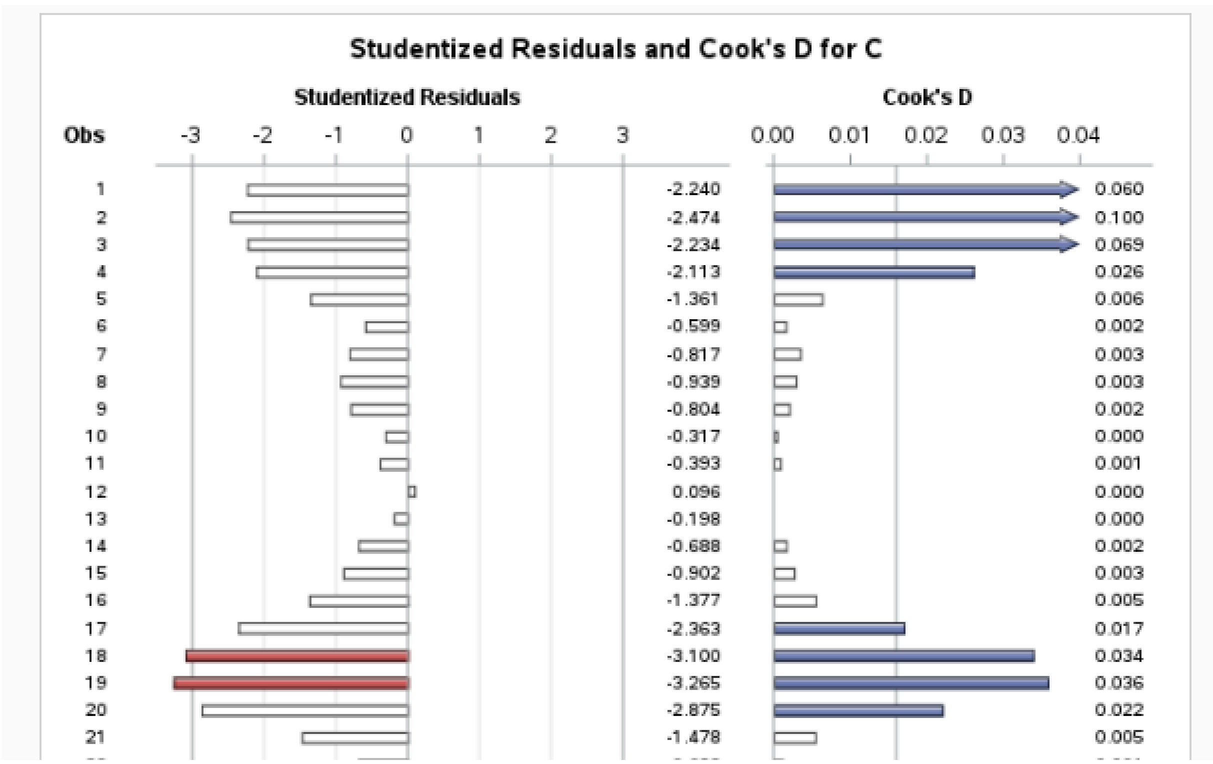


Figure 9: Cook's D and Studentized Residuals

The studentized residual plot shows observations 18 and 19 with large negative residuals close to -3. These points stand out slightly but are not extreme outliers. The Cook's D plot shows that observations 1, 2 and 3 have heavy influence on the model, meaning it may be affecting the regression results more than other data points.

Least Squares Prediction Equation:

Predicted CitiGroup Price = 36.54 + 5.367(BAC) - 0.386(COF) - 3.752(BK) + 0.139(BLK) - 0.0137(BK_SQ) + 0.0274(BAC_COF) + 0.046(BAC_BK) - 0.01418(BAC_BLK) - 0.00725(COF_BK) + 0.00580(BK_BLK)

- Interpretation of β_0: The model predicts CitiGroup's stock price to be $36.54, if all the other variables were zero.
- Interpretation of β_1 (BAC): For each $1 increase in Bank of America's stock price, CitiGroup's price is expected to increase by about $5.37, holding all other variables and interactions constant.
- Interpretation of β_2 (COF): For each $1 increase in Capital One's stock price, CitiGroup's price is expected to decrease by about $0.39, holding other variables constant.
- Interpretation of β_3 (BK): For each $1 increase in Bank of New York Mellon's stock price, CitiGroup's price is expected to decrease by about $3.75, holding other predictors constant.
- Interpretation of β_4 (BLK): For each $1 increase in BlackRock's stock price, CitiGroup's price is expected to increase by about $0.14, holding all other predictors constant.
- Interpretation of β_5 (BK_SQ): As BK increases, its impact on CitiGroup grows at first but then begins to slow down and level off. In other words, the relationship is curved rather than a straight line, and changes in BK have a smaller effect on CitiGroup at higher BK values.
- Interpretation of β_6 (BAC_COF): For each 1-unit increase in the product of Bank of America and Capital One's prices, CitiGroup's price increases by $0.027, meaning BAC's effect depends partly on COF's value.
- Interpretation of β_7 (BAC_BK): For each unit increase in the interaction between Bank of America and Bank of NY Mellon, CitiGroup's price increases by $0.046.

- Interpretation of β_8 (BAC_BLK): This coefficient is negative (–$0.014), meaning at higher BLK values, the positive effect of BAC on CitiGroup becomes slightly smaller.

- Interpretation of β9 (COF_BK): This interaction is negative (–$0.00725), indicating that when Capital One and Bank of NY Mellon increase together, CitiGroup's price decreases slightly.
- Interpretation of β10 (BK_BLK): This positive interaction ($0.0058) shows that Bank of NY Mellon and BlackRock strengthen each other's effect on CitiGroup's price.

Final Conclusions and Next Steps

The overall results of this analysis show that CitiGroup's daily stock price can be predicted very accurately using a small set of financial sector variables. Starting with 74 potential predictors, correlation analysis and stepwise regression helped narrow the model to a focused group of stocks with strong statistical relationships to CitiGroup. After addressing multicollinearity and testing both interaction and quadratic terms, the final model selected included Bank of America, Capital One, Black Rock, and Bank of New York Mellon, along with a quadratic term for Bank of NYM and significant interaction terms. This model explained 98.34% of the variation in CitiGroup's price and demonstrated low prediction error and stable coefficient estimates.

One important finding of the project was that many financial stocks move together, especially large banks and consumer finance companies. This created multicollinearity in the early stages of modeling and required careful removal of variables to ensure a good predictive model. The diagnostic plots confirmed that the final model met the key assumptions of a good regression model, with no major concerns in the residual patterns, but some signs of influential outliers were present in the Cook's D plot.

If this analysis were extended, a useful next step might be to include broader market variables, such as interest rates or economic indicators, since they may help explain some of the overall movement seen in the financial sector. Overall, the final model provides a strong starting point, and with additional data or methods, it could be improved even further.

Appendix

SAS Code:

```
FILENAME REFFILE '/home/u63886066/Lab 3/Fall 2025 Lab #3 Closing Prices.xlsx';

PROC IMPORT DATAFILE=REFFILE
    DBMS=XLSX
    OUT=WORK.stock;
    GETNAMES=YES;
RUN;

PROC CONTENTS DATA=WORK.stock; RUN;


%web_open_table(WORK.IMPORT);

proc print data = stock; run;

proc corr data=stock nosimple;
    var C;
    with _numeric_;
run;
```

```
proc reg data = stock;
    model C = WRB WFC TRV TFC SYF STT SCHW RJF RF NTRS NDAQ MS MMC MCO MA L KEY
    JPM JKHY IVZ IBKR HOOD HIG HBAN GS GL FISV FDS ERIE COIN COF CINF CFG CBOE BRO
    BLK BK BEN BAC AXP/selection = stepwise;
    run;

proc corr data=stock
    plots(maxpoints=none)=scatter(ellipse=none);
    var GS JPM BK IBKR MS HOOD COF NDAQ BAC BEN AXP SYF GL ERIE MCO HIG TRV WRB BLK;
    with C;
run;

proc corr data=stock
    plots(maxpoints=none)=scatter(ellipse=none);
    var HOOD COF NDAQ BAC BEN;
    with C;
run;

proc corr data=stock
    plots(maxpoints=none)=scatter(ellipse=none);
    var AXP SYF GL ERIE MCO;
    with C;
run;

proc corr data=stock
    plots(maxpoints=none)=scatter(ellipse=none);
    var HIG TRV WRB BLK;
    with C;
run;
```

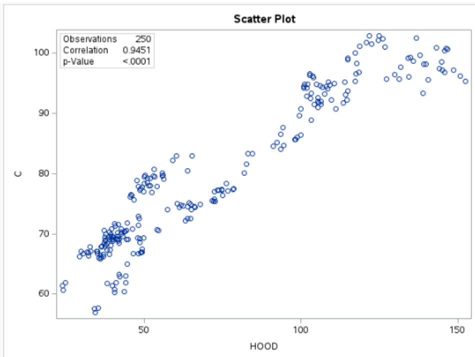
```
proc reg data = stock;
    model C = AXP GS IBKR MS HOOD JPM BAC COF BEN BK NDAQ BLK/selection = stepwise;
    run;

proc glmselect data = stock plots = criterionpanel;
    model C = AXP GS IBKR MS HOOD JPM BAC COF BEN BK NDAQ BLK/selection = stepwise stats = all;
    run;

proc reg data = stock;
    model C = GS IBKR MS BAC COF BEN BK BLK/vif;
    run;

proc reg data = stock;
    model C = IBKR MS BAC COF BEN BK BLK/vif;
    run;

proc reg data = stock;
    model C = IBKR BAC COF BEN BK BLK/vif;
    run;

proc reg data = stock;
    model C = BAC COF BEN BK BLK/vif;
    run;

proc reg data = stock;
    model C = BAC COF BK BLK/vif;
    run;

proc reg data = stock plots = (all diagnostics (unpack));
    model C = BAC COF BK BLK/all;
    run;
```

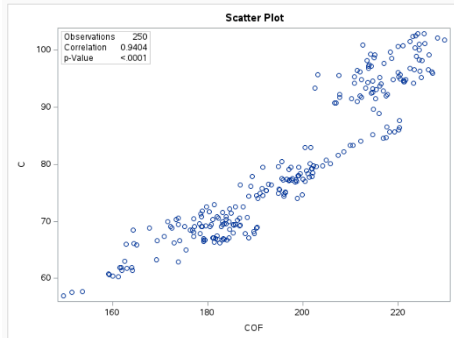
```
proc glm data=stock;
    model C = BAC | COF | BK | BLK @2 / solution;
    store GLMMODEL;
run;

proc plm restore=GLMMODEL noinfo;
    /* 1. Interactions with BAC */
    effectplot slicefit(x = BAC sliceby = COF);
    effectplot slicefit(x = BAC sliceby = BK);
    effectplot slicefit(x = BAC sliceby = BLK);

    /* 2. Interactions with COF*/
    effectplot slicefit(x = COF sliceby = BK);
    effectplot slicefit(x = COF sliceby = BLK);

    /* 3. Interaction between BK and BLK */
    effectplot slicefit(x = BK  sliceby = BLK);
run;
```

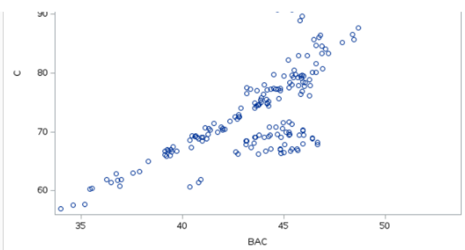
```
data stock2;
    set stock;

    BAC_COF = BAC * COF;
    BAC_BK  = BAC * BK;
    BAC_BLK = BAC * BLK;
    COF_BK  = COF * BK;
    COF_BLK = COF * BLK;
    BK_BLK  = BK  * BLK;
    BAC_SQ = BAC**2;
    BK_SQ = BK**2;
    COF_SQ = COF**2;
    BLK_SQ = BLK**2;
    RUN;

proc reg data = stock2;
    model C = BAC COF BK BLK BAC_COF BAC_BK BAC_BLK COF_BK COF_BLK BK_BLK;
    run;

proc reg data = stock2;
    model C = BAC COF BK BLK BAC_COF BAC_BK BAC_BLK COF_BK BK_BLK;
    run;

proc reg data = stock2;
    model C = BAC COF BK BLK BAC_COF BAC_BK BAC_BLK COF_BK BK_BLK BK_SQ;
    run;

proc reg data = stock2;
    model C = BAC COF BK BLK;
    run;

proc reg data = stock2 plots = (all diagnostics (unpack));
    model C = BAC COF BK BLK BAC_COF BAC_BK BAC_BLK COF_BK BK_BLK BK_SQ/all;
    run;
```

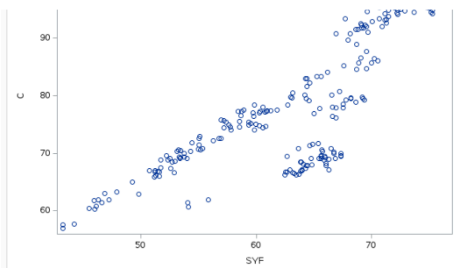
SAS OUTPUT:

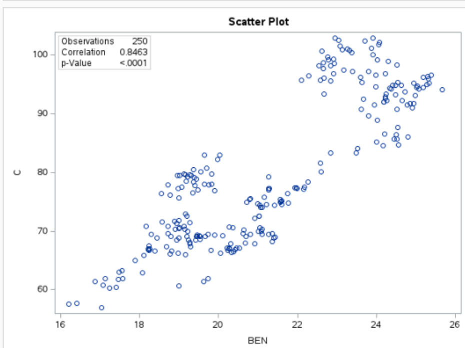
Summary of Stepwise Selection									
Step	Variable Entered	Variable Removed	Label	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	GS		GS	1	0.9700	0.9700	1410.50	8005.47	<.0001
2	BK		BK	2	0.0113	0.9812	791.567	148.10	<.0001
3	TRV		TRV	3	0.0070	0.9882	406.419	146.88	<.0001
4	COF		COF	4	0.0014	0.9896	332.653	32.42	<.0001
5	GL		GL	5	0.0014	0.9910	258.321	37.53	<.0001
6	HIG		HIG	6	0.0005	0.9915	231.326	15.08	0.0001
7	SYF		SYF	7	0.0004	0.9919	211.255	12.00	0.0006
8	JPM		JPM	8	0.0009	0.9928	162.858	30.76	<.0001
9		TRV	TRV	7	0.0000	0.9928	160.980	0.07	0.7851
10	MS		MS	8	0.0004	0.9933	139.045	15.55	0.0001
11	BEN		BEN	9	0.0003	0.9936	122.786	12.42	0.0005
12	BAC		BAC	10	0.0002	0.9938	114.909	6.88	0.0093
13	HOOD		HOOD	11	0.0003	0.9940	102.789	10.22	0.0016
14	WRB		WRB	12	0.0003	0.9944	87.3092	13.31	0.0003
15	IBKR		IBKR	13	0.0002	0.9946	75.8310	10.68	0.0012
16	MCO		MCO	14	0.0002	0.9948	69.2741	6.95	0.0089
17	ERIE		ERIE	15	0.0002	0.9949	60.7075	8.87	0.0032
18	AXP		AXP	16	0.0002	0.9951	53.0135	8.40	0.0041
19	NDAQ		NDAQ	17	0.0002	0.9953	45.1425	8.84	0.0033
20		GS	GS	16	0.0000	0.9953	45.0237	1.68	0.1957
21	WFC		WFC	17	0.0001	0.9954	41.6316	4.89	0.0279
22	CINF		CINF	18	0.0001	0.9954	40.6446	2.73	0.0998
23	BLK		BLK	19	0.0001	0.9955	37.9907	4.32	0.0389
24	STT		STT	20	0.0001	0.9956	34.0972	5.57	0.0191
25	GS		GS	21	0.0001	0.9957	32.5184	3.42	0.0657
26	HBAN		HBAN	22	0.0001	0.9957	31.3348	3.07	0.0811
27		HIG	HIG	21	0.0000	0.9957	30.1092	0.75	0.3884
28	COIN		COIN	22	0.0000	0.9958	29.7440	2.30	0.1310

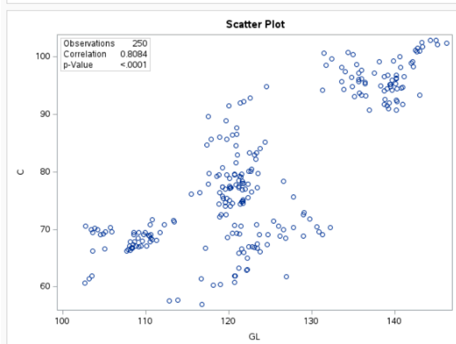

Scatter Plot
Observations 250
Correlation 0.9451
p-Value <.0001
C
HOOD

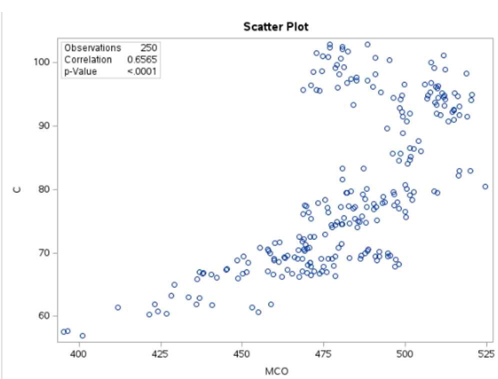

Scatter Plot
Observations 250
Correlation 0.9404
p-Value <.0001
C
COF

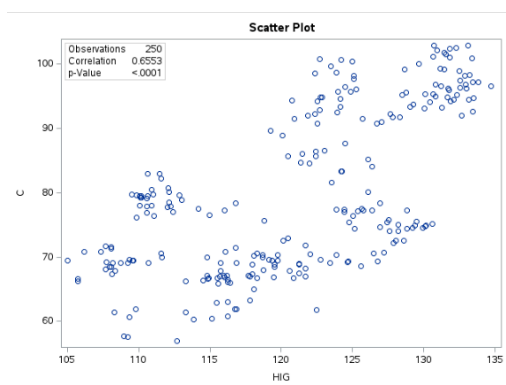

C
BAC

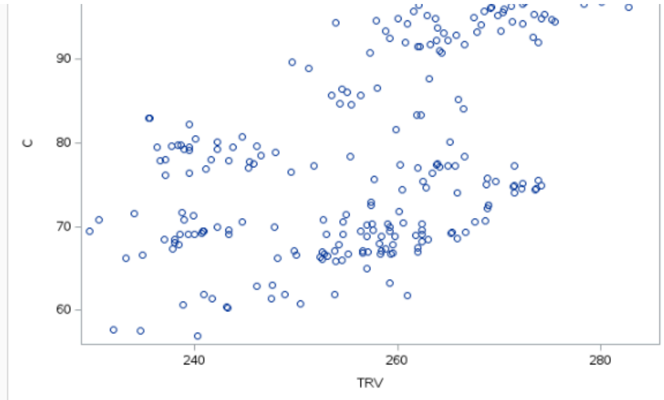

C
SYF

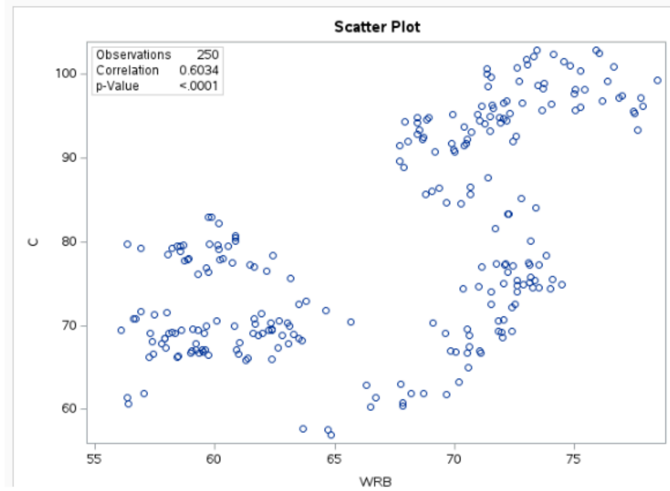

Scatter Plot
Observations 250
Correlation 0.8463
p-Value <.0001
C
BEN

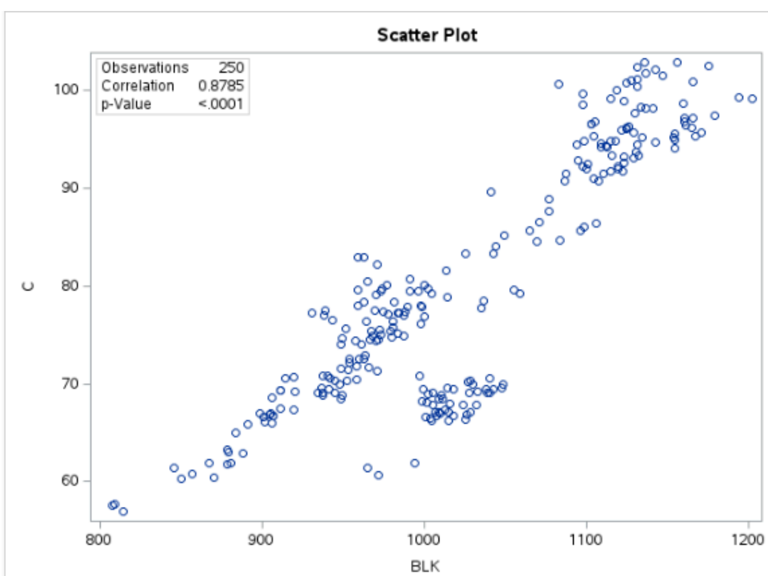

Scatter Plot
Observations 250
Correlation 0.8084
p-Value <.0001
C
GL


Scatter Plot
Observations 250
Correlation 0.6565
p-Value <.0001
C
MCO


Scatter Plot
Observations 250
Correlation 0.6553
p-Value <.0001
C
HIG


C
90
80
70
60
240
260
280
TRV


Scatter Plot
Observations 250
Correlation 0.6034
p-Value <.0001
C
100
90
80
70
60
55
60
65
70
75
WRB


Scatter Plot
Observations 250
Correlation 0.8785
p-Value <.0001
C
100
90
80
70
60
800
900
1000
1100
1200
BLK

(BLK)

Summary of Stepwise Selection									
Step	Variable Entered	Variable Removed	Label	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	GS		GS	1	0.9700	0.9700	384.612	8005.47	<.0001
2	BK		BK	2	0.0113	0.9812	150.229	148.10	<.0001
3	COF		COF	3	0.0011	0.9823	128.579	15.70	<.0001
4	BEN		BEN	4	0.0021	0.9845	86.0174	33.49	<.0001
5	MS		MS	5	0.0013	0.9858	60.6784	22.33	<.0001
6	BLK		BLK	6	0.0015	0.9873	31.0130	28.82	<.0001
7	BAC		BAC	7	0.0005	0.9878	22.1905	10.22	0.0016
8	IBKR		IBKR	8	0.0007	0.9885	9.0494	15.14	0.0001
9	NDAQ		NDAQ	9	0.0001	0.9887	8.1201	2.95	0.0870

The GLMSELECT Procedure
Selected Model

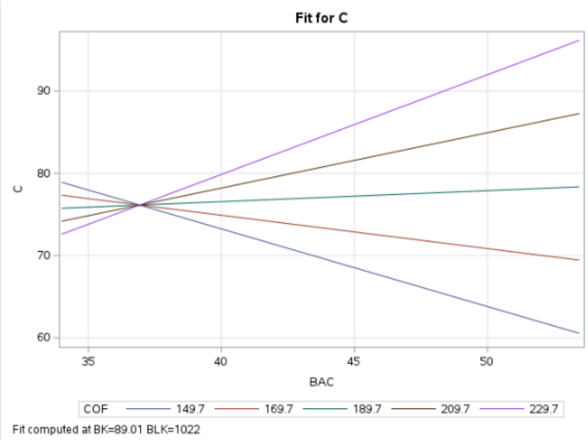
The selected model is the model at the last step (Step 8).

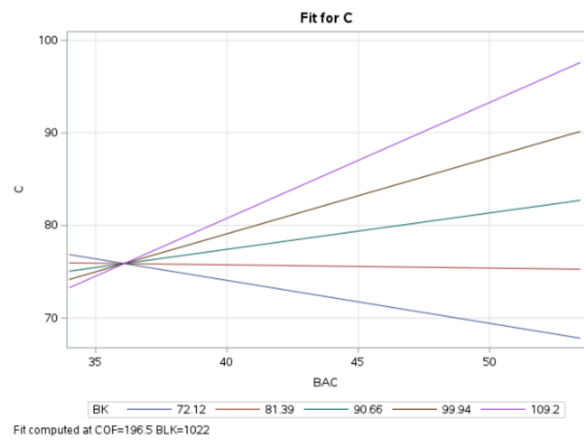
Effects:	Intercept GS IBKR MS BAC COF BEN BK BLK

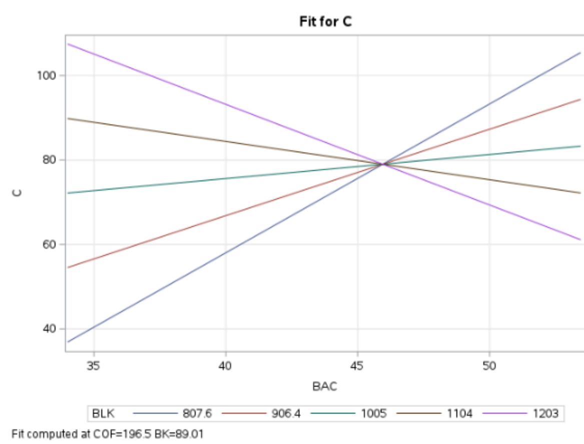
Analysis of Variance				
Source	DF	Sum of Squares	Mean Square	F Value
Model	8	38260	4782.53565	2592.69
Error	241	444.55421	1.84462	
Corrected Total	249	38705		

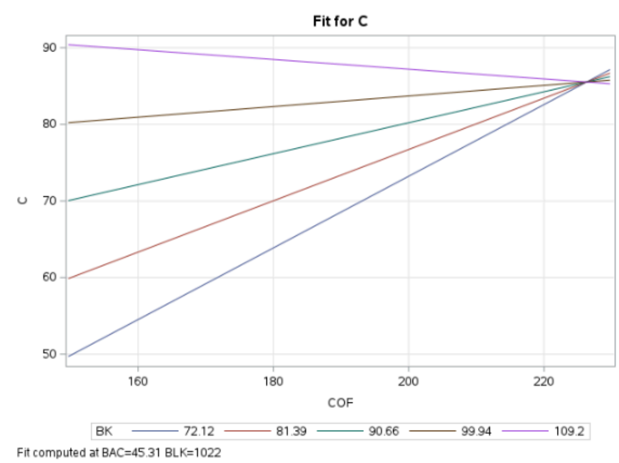
Root MSE	1.35817
Dependent Mean	79.72816
R-Square	0.9885
Adj R-Sq	0.9881
AIC	413.90277
AICC	414.82327
BIC	164.56839
C(p)	9.04941
PRESS	480.59145
SBC	193.59592
ASE	1.77822

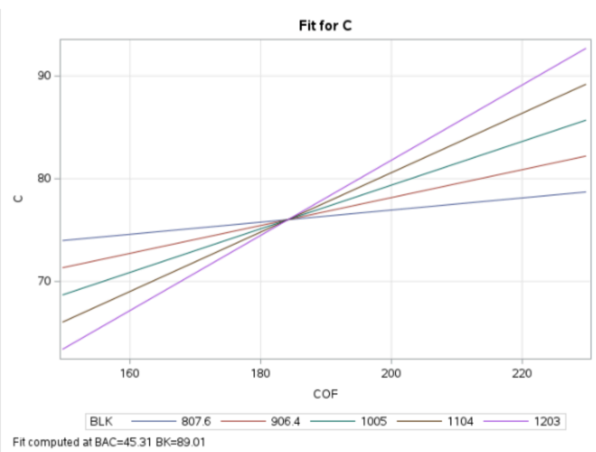
Parameter Estimates				
Parameter	DF	Estimate	Standard Error	t Value
Intercept	1	-20.580362	2.371127	-8.68
GS	1	0.093803	0.008294	11.31
IBKR	1	0.148621	0.038198	3.89
MS	1	-0.379686	0.047830	-7.94
BAC	1	0.459359	0.108285	4.24
COF	1	0.154636	0.018056	8.56
BEN	1	-1.066581	0.109316	-9.76
BK	1	0.423155	0.031370	13.49
BLK	1	0.016273	0.003435	4.74

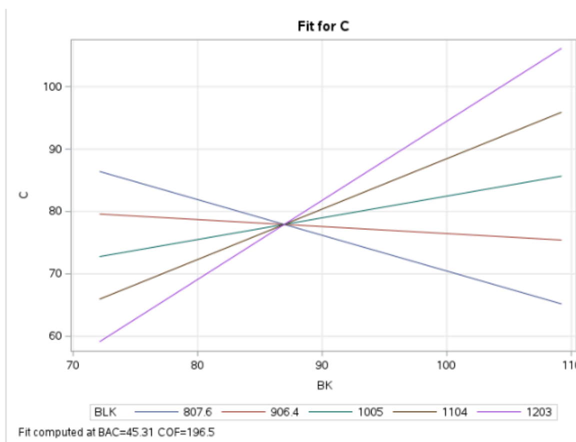

Fit for C
C
BAC
COF 149.7 169.7 189.7 209.7 229.7
Fit computed at BK=89.01 BLK=1022


Fit for C
C
BAC
BK 72.12 81.39 90.66 99.94 109.2
Fit computed at COF=196.5 BLK=1022


Fit for C
C
BAC
BLK 807.6 906.4 1005 1104 1203
Fit computed at COF=196.5 BK=89.01


Fit for C
C
COF
BK 72.12 81.39 90.66 99.94 109.2
Fit computed at BAC=45.31 BLK=1022


Fit for C
C
COF
BLK 807.6 906.4 1005 1104 1203
Fit computed at BAC=45.31 BK=89.01

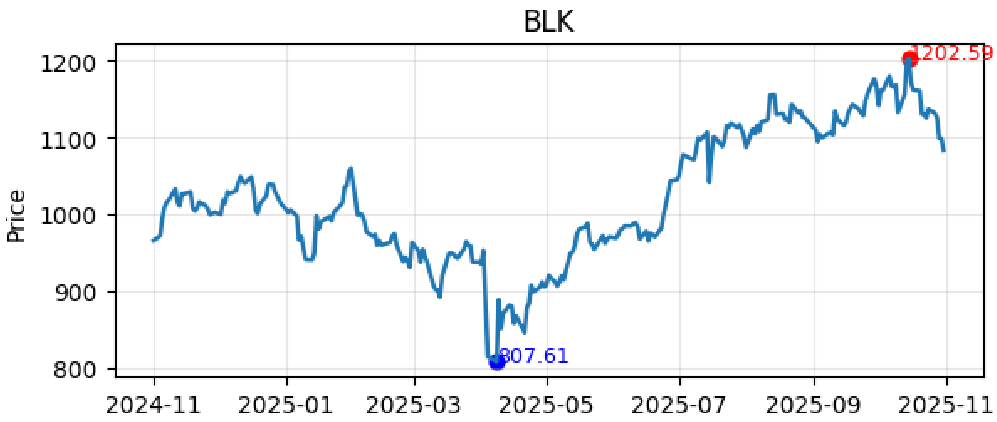

Fit for C
C
BK
BLK 807.6 906.4 1005 1104 1203
Fit computed at BAC=45.31 COF=196.5

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	10	38090	3808.96199	1479.70	<.0001
Error	239	615.21945	2.57414		
Corrected Total	249	38705			

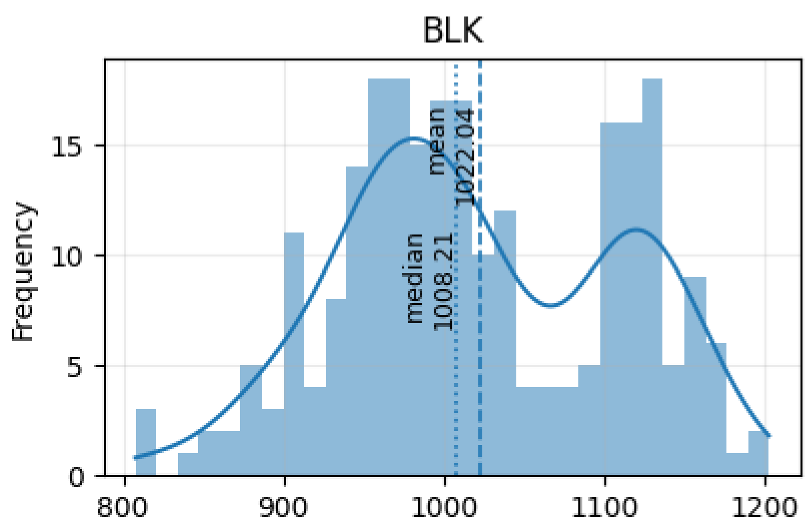
Root MSE	1.60441	R-Square	0.9841
Dependent Mean	79.72816	Adj R-Sq	0.9834
Coeff Var	2.01235		

Parameter Estimates						
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	Intercept	1	36.54318	17.04796	2.14	0.0331
BAC	BAC	1	5.36717	0.84267	6.37	<.0001
COF	COF	1	-0.38616	0.17367	-2.22	0.0271
BK	BK	1	-3.75190	0.55111	-6.81	<.0001
BLK	BLK	1	0.13946	0.04711	2.96	0.0034
BAC_COF		1	0.02746	0.00737	3.73	0.0002
BAC_BK		1	0.04628	0.01426	3.25	0.0013
BAC_BLK		1	-0.01418	0.00162	-8.76	<.0001
COF_BK		1	-0.00725	0.00314	-2.31	0.0219
BK_BLK		1	0.00580	0.00055304	10.49	<.0001
BK_SQ		1	-0.01374	0.00578	-2.38	0.0183

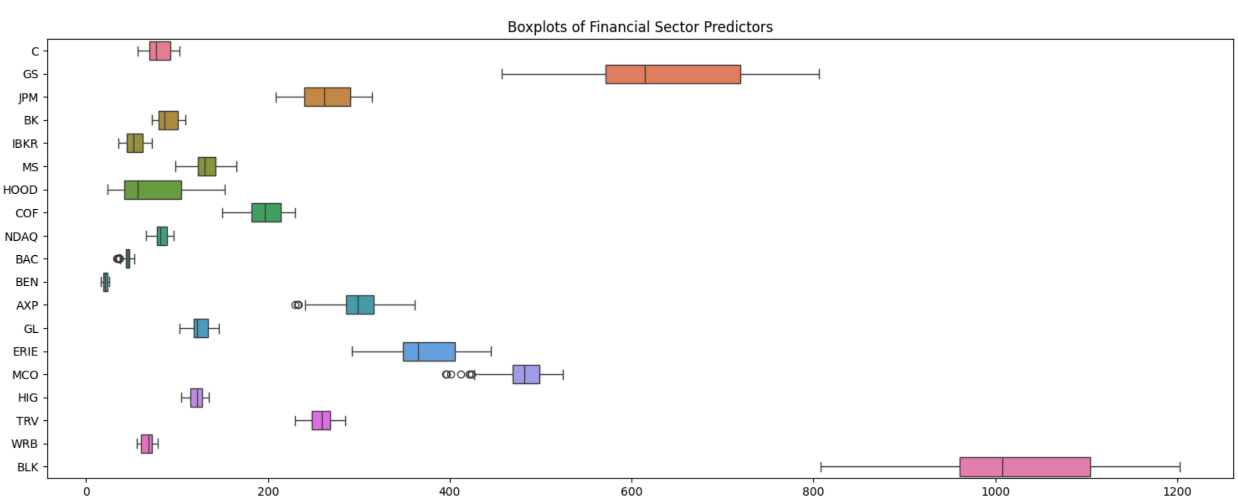
EDA



BLK
Price
1200
1100
1000
900
800
1202.59
807.61
2024-11
2025-01
2025-03
2025-05
2025-07
2025-09
2025-11



BLK
Frequency
mean
1022.04
median
1008.21
0
5
10
15
800
900
1000
1100
1200



Boxplots of Financial Sector Predictors
C
GS
JPM
BK
IBKR
MS
HOOD
COF
NDAQ
BAC
BEN
AXP
GL
ERIE
MCO
HIG
TRV
WRB
BLK
0
200
400
600
800
1000
1200